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COLLEGE OF NATURAL AND COMPUTATIONAL SCIENCE

DEPARTMENT OF INFORMATION SYSTEM

Object Oriented System Analysis and Design

Dormitory Housing Management System

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1. Select Requirement Gathering Techniques

Technique 1: Interviews

Interviewing a dorm proctor and security guard to confirm real workflows; how rooms are assigned, how penalties are recorded, and how laundry exits are managed.

Interviews provide deep, qualitative insights.

Technique 2: Survey

Surveying students to priorities features. For example, how often do they need maintenance, whether they want a QR-based laundry system.

Surveys reach many stakeholders quickly.

Technique 3: Document Analysis

Analyzing existing documents: complaint registers, penalty logbooks, room allocation sheets, and duty.

This helps in understanding the current process and identify gaps.

Technique 4: Brainstorming

In a team workshop, we will brainstorm “blue-sky” ideas such as automatic room assignments, push notifications to staff, or a penalty appeal system.

Brainstorming encourages creative requirements.

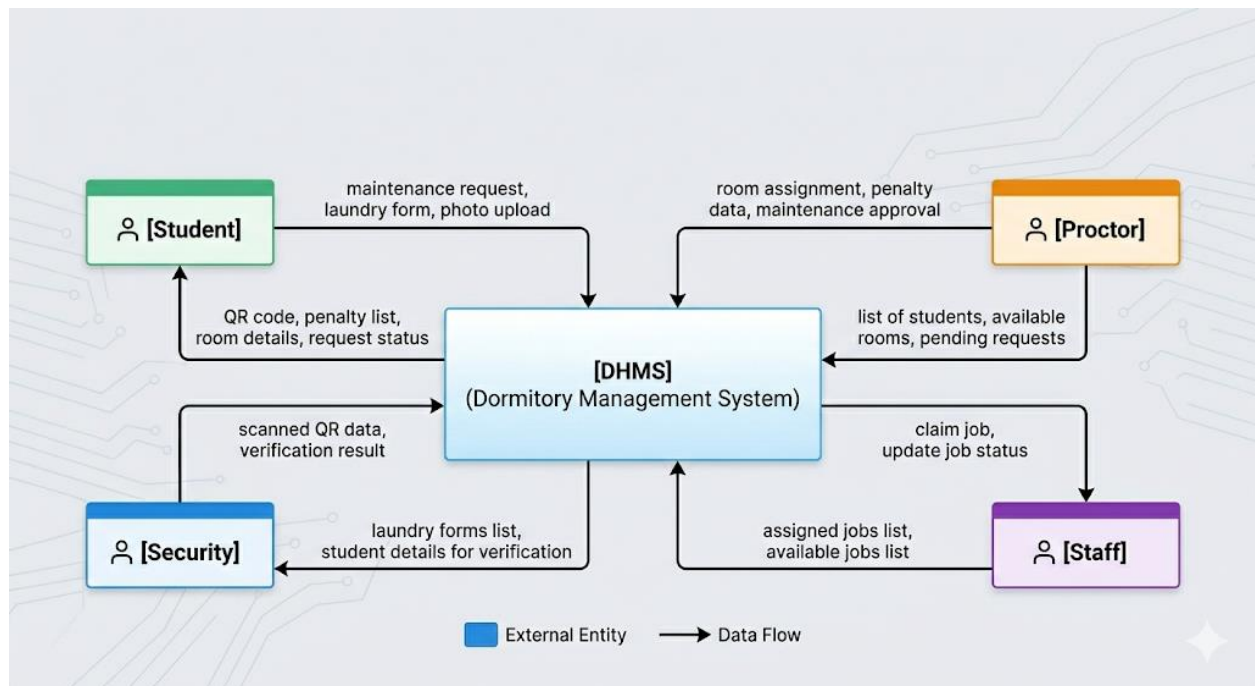
2. Essential Use Case Modeling

1. Use Case Summary

Use Case Name	Primary Actor	Description (user goal)
Assign Room to Student	Proctor	The proctor selects a student, a dormitory, a room number, an assignment date, and an expected check-out date.
View Available Rooms	Proctor	The proctor views a list of all rooms with details (dorm, floor, type, capacity, occupancy, status).
Assign Penalty	Proctor	The proctor selects a student, a violation type, description, duration, and start date; the system calculates the end date.
Approve Maintenance Request	Proctor	The proctor reviews pending requests and assigns them to appropriate staff.
Submit Maintenance Request	Student	The student describes an issue (type, title, description) and sets an urgency level.
Submit Laundry Form	Student	The student enters number of items, item list, and special instructions; the system generates a QR code for exit.
Verify Laundry Forms	Security	The security guard views a list of submitted laundry forms and marks them as verified.

Scan Laundry QR Code	Security	The security guard scans the student's QR code using a camera to verify the laundry exit.
View My Assigned Jobs	Staff	The staff member sees all jobs assigned to them (ID, room, issue, urgency, status).
Claim Available Job	Staff	The staff member browses unassigned jobs and claims one matching their skill.

2. Context Diagram



3. Primary and Secondary Actors

Actor	Type	Description
Student	Primary	A resident who uses the system to request maintenance, submit laundry forms, and view penalties.
Proctor	Primary	A dormitory supervisor who assigns rooms, approves maintenance requests, assigns penalties, and views student lists.
Staff	Primary	Technical staff who receive assigned jobs, update job status, and claim available tasks.
Security	Primary	A guard who verifies laundry exit QR codes and manages student check-in/out.
Notification Service	Secondary	An external system (email/SMS) that sends alerts to students and staff.

4. Requirements Anticipation, Elicitation, Assurance, Specification

A. Requirements Anticipation: Predicting what users will need without being explicitly asked, based on domain knowledge and experience.

- We anticipate that students will want to track the status of their maintenance request (e.g., “pending → assigned → completed”) even though no student has mentioned it yet. We will include a status field in the requirements.

B. Requirements Elicitation: Actively gathering requirements from stakeholders through interviews, surveys, workshops, observation, etc.

- We will elicit requirements by showing a mock-up of the penalty form to a proctor and asking: “What violation types should be included? Do you need a ‘consequences’ field?”.

C. Requirements Assurance: Ensuring that requirements are correct, consistent, feasible, and complete by using validation techniques such as inspections, walkthroughs, and prototyping.

- We will perform assurance by reviewing all requirements with our team and a proctor. We will check that “the system shall generate a QR code for laundry” is consistent with “security shall scan QR code” and is feasible using open-source libraries.

D. Requirements Specification: Writing requirements clearly and unambiguously in a document (e.g., Software Requirements Specification).

- Example specification: “**The DHMS shall allow a student to submit a maintenance request by providing issue type (plumbing, electrical, furniture, other), a brief title, a detailed description, and an urgency level (low, medium, high).**” This will be documented in our SRS.

3. Activity Diagram

1. Additional use cases missed earlier

After reviewing the dormitory management domain again, we identified the following additional use cases that were not included in our original summary:

New Use Case	Actor	Reason for adding
Update Student Profile Photo	Student	Required for identification; previously implied but not listed.

Notify Staff of New Job	System (Notification Service)	Automatic alert when a maintenance request is approved and assigned.
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Revised total major use cases: 12 (added the two above to the original ten).

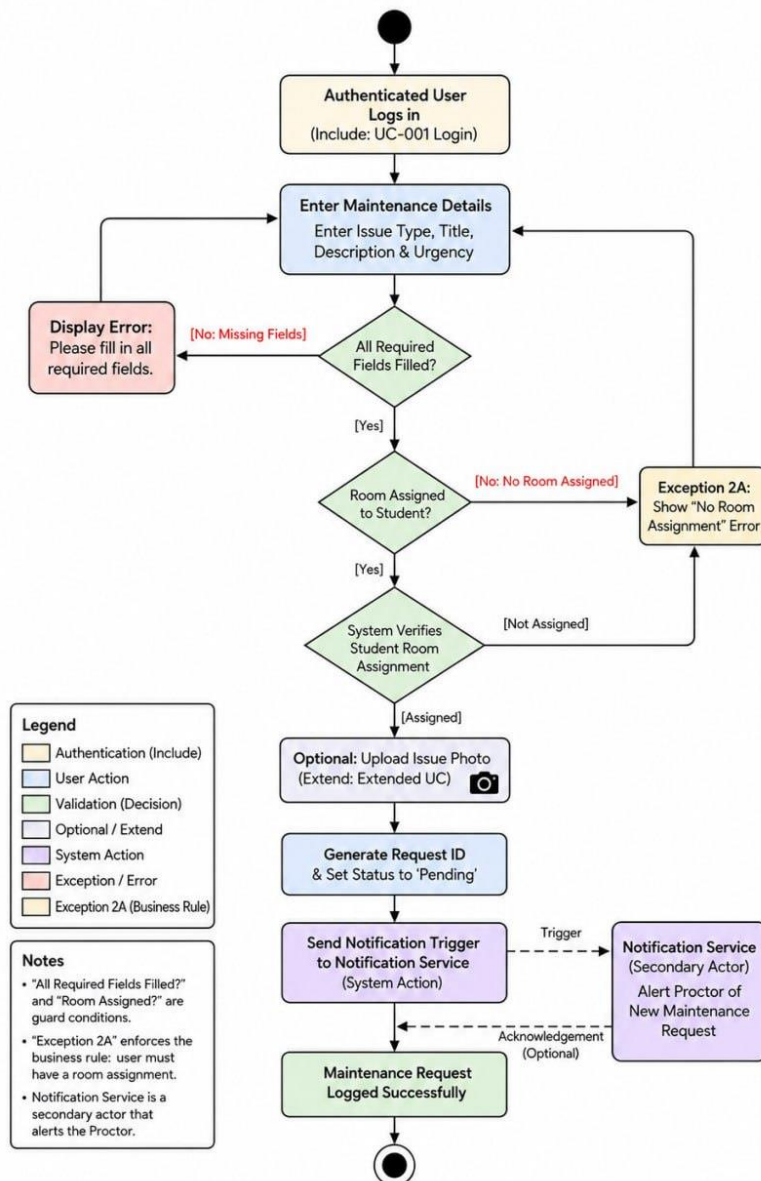
2. Use case template

Below is a **use case template** filled for one of our DHMS use cases. This template can be reused for all other use cases.

Element	Description
Use Case Name	Submit Maintenance Request
Use Case ID	UC-003
Primary Actor	Student
Secondary Actors	Proctor (for approval), Staff (for assignment)
Goal	Student reports a problem in their dorm room so that it can be fixed.
Precondition	Student is logged in and has an assigned room.
Trigger	Student clicks “Maintenance” in the dashboard and selects “New Request”.
Normal Flow	1. Student selects issue type (plumbing/electrical/furniture/other). 2. Student enters a brief title. 3. Student enters a detailed description. 4. Student selects urgency (low/medium/high). 5. Student clicks “Submit”. 6. System saves the request with status “Pending”. 7. System notifies the proctor (optional).
Alternative Flows	4A. Student may also upload a photo (future enhancement). 6A. If required fields missing, system prompts to complete.
Exceptions	2A. Student not assigned a room → system shows error: “No room assigned. Contact proctor.”

Postcondition	Maintenance requests are stored in the system with a unique ID and pending status.
Assumptions	Internet connection is available. Student has valid login.

UML Activity Diagram: UC-003 Submit Maintenance Request



Description of Activity Diagram (UC-003): The diagram above illustrates the workflow for submitting a maintenance request. It begins with user authentication (**Login**) and proceeds to data entry. The system performs two critical validation checks: ensuring all required fields are populated and verifying that the student has an active **Room Assignment**. If these conditions are met, the request is logged, an optional photo can be attached (**Extend** relationship), and the **Notification Service** alerts the relevant Proctor.

3. Generalize actors if necessary

No actor generalization is needed for this DHMS.

- All actors (Student, Proctor, Staff, Security) have distinct roles and responsibilities.
- Staff includes plumbers, electricians, etc. But they are **not separate actors** – they all share the same use cases (view assigned jobs, claim available jobs, update status). Therefore, we treat **Staff** as a single actor.
- No parent-child relationship exists between actors in our system.

4. Include and Extend use cases

- **Include relationships:** base use case depends on included use case.

Base Use Case	Included Use Case	Reason
Submit Maintenance Request and Laundry Form	Login	Students must be authenticated before submitting.
Assign Penalty	View Student List	Proctor must see the list of students before selecting one.

Verify Laundry Form	Scan QR Code	Security must scan the QR code as part of verification.
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- **Extend relationships:** extending use case adds optional behaviors.

Base Use Case	Extending Use Case	Extension Condition
Submit Maintenance Request	Upload Photo (optional)	If the student chooses to attach a photo of the issue (future enhancement).
Assign Penalty	Send Appeal Notification	Only if the penalty duration exceeds 7 days (future enhancement).
Scan QR Code (Security	Print Verified Slip	If security wants to print a receipt after verification.

5. Generalize use cases if necessary

Given the size of the DHMS, **use case generalization is not necessary**. We can keep all use cases at the same level without a parent-child hierarchy.